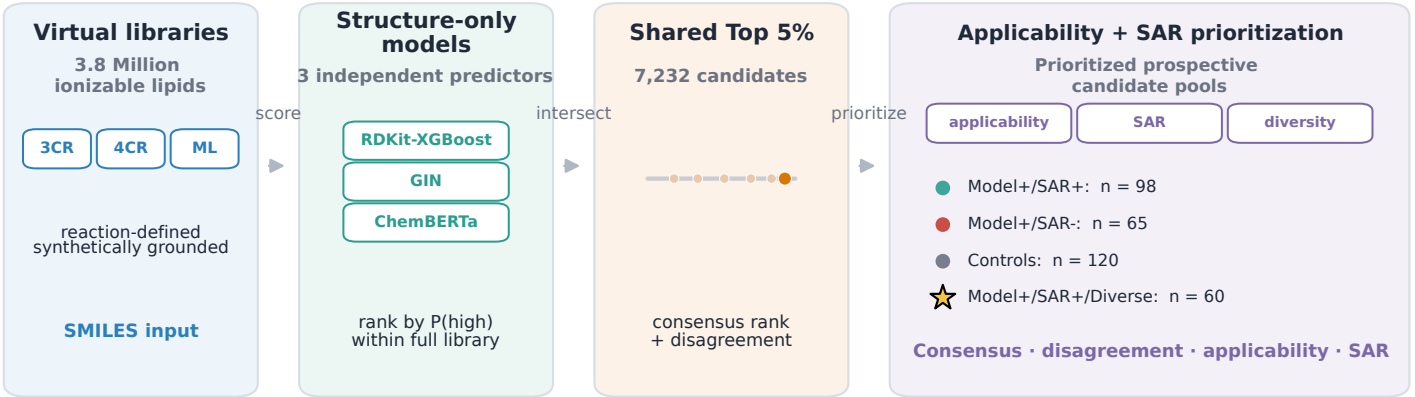
Virtual libraries
3.8 Million
ionizable lipids
3CR
4CR
ML
reaction-defined
synthetically grounded
SMILES input
score
Structure-only
models
3 independent predictors
RDKit-XGBoost
GIN
ChemBERTa
rank by P(high)
within full library
intersect
Shared Top 5%
7,232 candidates
consensus rank
+ disagreement
prioritize
Applicability + SAR prioritization
Prioritized prospective
candidate pools
applicability
SAR
diversity
Model+/SAR+: n = 98
Model+/SAR-: n = 65
Controls: n = 120
Model+/SAR+/Diverse: n = 60
Consensus · disagreement · applicability · SAR